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Task Description:

Knowledge of capillary flows in complex geometries is fundamental to many space fluids systems designs. Such knowledge provides a basis to exploit passive means of containment and management for large liquid inventories aboard spacecraft. Such knowledge is also essential to prevent inadvertent destabilization and deactivation of critical systems where capillary forces can dominate fluid behavior. Complimentary but distinctly non-computationally-intensive methods to design and analysis such are in need by the aerospace community tasked to meet NASA's challenges for space exploration in the present and future. Specific progress is reported here for three fundamental and applied research projects: (1) the NRA investigation "Capillary Flow in Complex Geometries," (2) ISS Flight experiment results and analysis for the "Capillary Flow Experiment" (CFE), and (3) progress in the development of the German-US collaborative ISS experiment "Capillary Channel Flow" (CCF) experiment. All are briefly highlighted in this summary and addressed individually in greater detail in the subsequent report. Review of Objectives. Large length scale capillary flows in interior corners have an established place in fluids-handling operations in reduced gravity environments. Such flows are driven by wetting conditions and the often complicated geometry of the system. A quantitative and fundamental understanding of corner flows is essential for the myriad fluids management tasks aboard spacecraft including flows in liquid propellant and cryogen storage systems, life support systems, phase-change thermal systems for temperature control and power production, and for materials and biofluids processing. Recent advances in the analysis of capillary flows in containers with interior corners now permit the closed form prediction of flows in a variety of planar and nonplanar corners, 2-D corner networks, and 3-D containers such as cylinders with irregular polygonal sections and cylinders with complex vaned sections—models for current propellant management devices. The analytical solutions serve as design tools that can significantly reduce the design cycle time for low-g capillary systems and allow new insights into flow mechanisms leading to designs with improved performance and reliability without a mass penalty.

In this investigation, analytical solutions of large length scale capillary driven flows in systems of increased geometric complexity are pursued: these include flows within systems of non-idealized corners, bifurcating and coalescing flows in corner networks, multidimensional imbibition, and flows in systems with partial wettability. Such flows are unique to the low-g environment, commonplace in spacecraft fluid systems, and advantageous if exploited, yet at times overlooked during system development. In this work, useful analytic expressions for key flow characteristics such as transient interface shape, flow rate, and stability are sought. Low-g aircraft and drop tower experiments are proposed to support the theoretical analyses. Though the proposed research is fundamental in terms of analysis and experimental verification, each problem to be addressed has broad application to fluids management in low-g and terrestrial environments. In addition to expected publications, the new results along with previous solutions will be collected and published in handbook form for use by engineers in the design of advanced fluids systems for spacecraft. 1. NRA-Research, Capillary Flow in Complex Geometries. This work has produced numerous design guides for passive fluid control aboard spacecraft. The work has also spun out methods to address microscale capillary flows in porous materials for terrestrial applications. Significant progress in the first year research effort was made. For example, a first thorough analysis for flows in rounded interior was completed and accepted for publication in the Journal of Fluid Mechanics. Drop tower experiments performed at NASA GRC were conducted to benchmark the theoretical predictions. Further drop tower experiments were performed to investigate flows in containers with variable wettability, 3-D corner networks, and corners possessing gaps—models of systems commonly employed in spacecraft propellant management schemes. The shapes and stability of large wall-edge-bound capillary drops and bubbles were computed and similar computations are underway for drops and bubbles on 3-D structures such as screens also employed in low-g fluid management applications. Two journal articles appeared in the literature concerning this research, one article was accepted for publication, and another is currently in preparation. A NASA technical memorandum was published and eight conference presentations were made, with many outreach opportunities exploited to introduce local area students to low-g physical phenomena, NASA-related research, ISS utilization, and the challenges of aerospace engineering. One M.S.M.E. student, one Post Doctoral Fellow, one visiting scholar (Bremen, Germany), two summer high school students, and a graduate course (ME510 Capillary Phenomena, 3 students) were in part supported by this research effort.

2. Capillary Flow Experiment (CFE): The CFE Contact Line-2 (CL-2) module was operated most recently on ISS on Tuesday, April 18, 2006, by astronaut Jeff Williams. The tests represent the fourth and final operation of the handheld fluids experiment previously conducted on ISS in August and September of 2004 by Mike Fincke, and in December 2005 by Bill McArthur. The objective of the CFE Contact Line experiment is to determine the impact of the moving contact line boundary condition on large capillary surfaces in space. The contact line boundary condition is an essential but often unknown quantity when predicting, for example, the behavior of large liquid inventories in spacecraft fuel/cryogen tanks. Nearly 200 data sets were collected for the CFE Contact Line-2 unit for a significant variety of conditions including lateral and axial excitations. Natural frequencies, damping rates, and critical interface stability conditions are determined from the digitized data for comparison to 'blind' numerical predictions that are presently underway. Five more CFE test articles are currently stowed on the shuttle for transport to ISS during the next scheduled Shuttle launch. This work has in part supported one M.S.M.E. student, one undergraduate student, one summer high school student, and one visiting scholar (Bremen, Germany).

3. Capillary Channel Flow (CCF): In this work a collaboration between German investigator Michael Dreyer (Univ. Bremen/ZARM) and U.S. investigator M. Weislogel (Portland State University) was established to guide the German hardware design and development for future experiments aboard the ISS (Glovebox). The experiments concern the performance and stability of forced capillary flows and their dependence on system geometry. Such flows are critical to advanced low-gravity fuels and cryogens management. The science requirements for the hardware to meet the US objectives are nearing completion. Anticipated space flight of CCF is expected several years from now.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Below is listed the accomplishments of the first year effort. These are followed by an outline of the goals for subsequent year(s). The volume and pace of this research are enabled by a generalized analytical technique that has been found to produce manifold closed form solutions—each solution of which enables further solutions to related problems of interest. For example, in this first year effort it was found that multi-parameter geometric scaling could be applied to capillary driven flows in rounded interior corners collapsing all geometric dependence into intricate but know analytic expressions. This result is encouraging and strengthens the likelihood of further analytic solutions in problems of increased geometric complexity. The method is currently being applied to other geometries in hopes of establishing a scaling protocol for such problems. It remains surprising how simply certain capillary driven flows on/in complex geometries can be understood and analyzed with only second order reliance on numerical data.
• Experiments and analyses for capillary driven flows along interior corners that are discontinuously wetted were completed. A modified coordinate rotation and scaling technique was employed to produce a narrowly banded coefficient that may be retroactively applied to all previous capillary solutions for capillary driven flows along interiors that possess different wettabilities on the opposing faces of the planar corner walls. This flow has particular application in containers consisting of differing materials. One journal article was published on this subject during this period, also one NASA Contract Report, and an M.S. Thesis (C.L. Nardin 2005). Six drop tower experiments representing 12 separate experiments were completed in NASA GRC's 2.2s drop tower facility.

• The problem of capillary flow along interior corners that are rounded is an important practical problem for spacecraft fluid system design. Through a careful selection of geometric length scales and through the introduction of a new geometric scaling parameter, the governing equations were reduced to convenient forms for both analytic and numeric solutions for all values of corner angle and roundedness. The scaling and analysis of the problem captures much of the intricate geometric dependence of the viscous resistance and significantly reduces the reliance on numerical data compared to several previous solution methods and the numerous subsequent studies that cite them. In general, three asymptotic regimes may be identified from the large second order nonlinear evolution equation: (i) the `sharp corner' regime, (ii) the narrow `rectangular section', and (iii) the `thin film' regime. Flows are observed to transition between regimes, or may exist essentially in a single regime depending on the system. Similarity solutions are possible to the full equation which is readily solved numerically. Approximate analytical solutions are also possible under the constraints of the three regimes which are clearly identified. The general analysis enables analytic solutions to many rounded corner flows and example solutions for steady flows, perturbed infinite columns, and flows with finite tip curvature were computed. This work has been accepted for publication in J. Fluid Mech. It has also been presented at 2 technical symposia. Experimental results to support the theoretical findings are provided by way of 38 drop tower tests performed at NASA GRC. This data is currently being reduced for a subsequent publication.

• The Surface Evolver algorithm was employed to determine the behavior of wall-bound drops and bubbles which are critical to many capillary natural and industrial processes. They are important to capillary flow in interior corners in that they are isolated from the primary flow path and require a degree of forcing to be moved. Key characteristics of such drops and bubbles include interface shape and stability for a variety of gravity levels and orientations. Significant solutions are in hand for axisymmetric pendent drops for a variety of uniform boundary conditions along the contact line with gravity acting normal to a planar wall. The special case of a wall-bound drop or bubble that is also pinned at an edge (i.e. a `wall-edge-bound' drop) was considered here where numerical solutions are obtained for interface shape and stability as functions of drop volume, contact angle, fluid properties, and uniform gravity vector. For a semi-infinite zero-thickness planar wall (plate), a critical contact angle was identified below which wall-edge-bound drops are always stable. The critical contact angle is computed as a function of the gravity vector. The numerical procedure, which makes no account for contact angle hysteresis, predicts that such wall-edge-bound drops are unconditionally unstable for any gravity field with a component that is tangent to the wall while inwardly normal to the edge. The numerical results are correlated simplifying the prediction of drop/bubble release conditions for a variety of terrestrial and microgravity fluids systems. One publication and one presentation and one presentation appeared on this subject.

• Many practical capillary systems involve containers/geometries that are tapered or possess weakly 3-D dependence. The imbibition problem in containers with interior corners is being extended to such cases where the taper of, for example, an irregular polygonal cross section provides particular design advantages in preferentially locating the liquid where desired. Passive large length scale capillary flow in such containers cannot be tested on the ground. The equations governing the process are known but have not been solved analytically to date because of a lack of experimental data identifying the appropriate boundary conditions for the flow problem. Early indications are that the overall pressure difference for the flow may actually be constant—even for fairly complex 3-D shapes—which would greatly simplify analyses. Experimental results will guide the analysis by providing the necessary boundary condition(s) as a function of container cross section and fill fraction. Six 2.2s drop tower tests at the NASA GRC facility were completed to this end and the data are currently being digitized for analysis. (The tests cells employed for these experiments were the engineering units for CFE to be described in greater detail in Section II.) A benchmarked theory can then be used to design and analyze capillary devices such as 3-D vane networks and tapered screen galleries for bubble-free collection and positioning of fuels for satellites, an important and outstanding problem for propellant management aboard spacecraft.

• Numerical and scale analyses have begun to address corner flows for nonwetting fluids and along corners separated by gaps or other vane structures. Concerning the latter, 70 successful NASA GRC 2.2s drop tower tests were performed to serve as the database from which to direct the analytic effort. The challenges arise by the addition of two geometric parameters (gap width and thickness) to the already crowded list including corner angle, contact angle, container size and shape (3-D). Numerical analyses have been performed to date using a MatLab-based program benchmarked by the CFD package STAR-CD.

Directions. In the upcoming year, most of the work reported above should be complete. New experiments concerning bifurcating flows, coalescing flows, and flows in 2-D and 3-D interior corner network systems will be initiated. These flows will exploit ground experiments where possible, but will likely require tests aboard NASA's low-g aircraft. Response times are not high enough to make sufficient use of NASA drop towers. A 3-D interior corner flow model will be developed for such flows. Constraints for the model (i.e. container saturation rates) will be identified via scaling analysis and numerical analysis for static configuration using the Surface Evolver algorithm. Flows where capillary, viscous and inertial forces are of competing orders will also be planned with the support of drop tower and aircraft experiments as needed (see CCF section III). All results useful in practical system design are being and will be collected for a 'handbook-style' NASA contract report surveying capillary flows in complex geometric vessels, conduits, and tanks.

II. Capillary Flow Experiment (CFE)

During this research effort the PI supported the continued development, crew training, crew procedures, in flight operations, and reporting for the Capillary Flow Experiment performed aboard the International Space Station. CFE is a suite of fluid physics flight experiments whose purpose are to investigate large length scale capillary flows and phenomena in low gravity. The CFE data to be obtained will be crucial to the Space Exploration Initiative, particularly as pertains to fluids management systems, such as fuels/cryogen storage systems, water collection and recycling, thermal control systems, and materials processing in the liquid state. NASA's current plans for exploration missions assume the use of larger liquid propellant masses than have ever flown on interplanetary missions. Under low gravity conditions, capillary forces can be exploited to control fluid orientation so that such large mission-critical systems perform more reliably.

CFE is a simple fundamental scientific study that can yield quantitative results from safe, low-cost, short time-to-flight, handheld fluids experiments. The experiments aim to provide results of critical interest to the capillary flow community that cannot be achieved in ground-based tests; such as tests to probe dynamic effects associated with a moving contact boundary condition, capillary driven flows in interior corner networks, and critical wetting phenomena in complex geometries. Specific applications of the results center on particular fluids challenges concerning propellant tanks. The knowledge gained will help spacecraft fluid systems designers increase system reliability, decrease system mass, and reduce overall system complexity.

CFE encompasses three experiments with two unique experimental units per experiment. There are multiple tests per experiment. Each of the experiments employs parametric ranges and test cell dimensions that cannot be achieved in ground-based experiments. All units use similar fluid injection hardware, have simple and similarly sized test chambers, and rely solely on video for highly quantitative data. Silicone oil will be used for these tests. Differences between units are primarily fluid properties, wetting conditions, and test cell geometry. The experiment procedures are simple and intuitive.

One of the six CFE units, the CFE Contact Line-2 (CL-2) module, was operated on ISS on Tuesday, April 18, 2006, by astronaut Jeff Williams. The tests represented the fourth and final operation of the handheld fluids experiment previously conducted on ISS in August and September of 2004 by Mike Fincke, and in December 2005 by Bill McArthur. The latest experiments were conducted to collect critical statistical data thought not possible during the original planning of the experiment. The experiments were enabled by the previous on-orbit observations of Fincke and McArthur and were conducted skillfully by Williams who discovered a novel and highly repeatable method to reset the experiment for indefinite repeatability. Williams quickly became proficient at 'resetting' the experiment, permitting over 30 minutes of extra science and completing the experiment ahead of schedule. The science team was surprised to collect twice the data it sought with additional data collected for important phenomena not included in the original science requirements.

The objective of the CFE Contact Line experiment (PI: M. Weislogel Portland State University) is to determine the impact of the moving contact line boundary condition on large capillary surfaces in space. The contact line boundary condition is an essential but often unknown quantity when predicting, for example, the behavior of large liquid inventories in spacecraft fuel/cryogen tanks. Williams successfully collected CFE Contact Line-2 data for the lateral and axial excitations of cylindrical containers, determined natural frequencies, damping rates, critical interface stability conditions, and determined drop coalescence and adherence limits applicable to the design of systems requiring droplet two-phase flow separations. The down-linked video will be digitized immediately for 'blind' numerical predictions that are presently underway for the previous experiments. The computations are being performed by a visiting researcher from the laboratory of Dr. M. Dreyer of the University of Bremen Germany and ZARM, inc.

Nearly 200 data sets were collected for the CFE Contact Line-2 unit for a significant variety of conditions including lateral and axial excitations. Natural frequencies, damping rates, and critical interface stability conditions are determined from the digitized data for comparison to 'blind' numerical predictions that are presently underway. Five more CFE test articles are currently stowed on the shuttle for transport to ISS during the next scheduled Shuttle launch. This work has in part supported one M.S.M.E. student, one undergraduate student, one summer high school student, and one visiting scholar (Bremen, Germany). Drop tower test at NASA GRC's 2.2s facility were conducted in support of the flight research. A total of 17 tests were conducted employing engineering models of all three CFE container types: Contact Line, Vane Gap, and Interior Corner Flow (3 CL, 8 VG, 6 ICF)

III. Capillary Channel Flow (CCF)

In this work a collaboration between German Principal investigator Michael Dreyer (Univ. Bremen/ZARM) and U.S. Co-investigator M. Weislogel (Portland State University) was established to guide the German hardware design and development for future experiments aboard the ISS (Glovebox). Weislogel assisted the project both to widen the science return of the experiments by providing several application specific geometries and test parameters. A second test cell is to be constructed to this end. In general, the experiments concern the performance and stability of forced capillary flows and their dependence on system geometry. Such flows are critical to advanced low-gravity fuels and cryogens management. The science requirements for the hardware to meet the US objectives are nearing completion. Anticipated space flight of CCF is expected several years from now and a letter of understanding between NASA and DLR is being prepared. To meet the stringent project deadlines the PI and Co-PI have met in Germany and twice in the U.S. to prepare documents, discuss science focus, and prepare technical presentations on the subject. Two technical presentations have resulted form this work during this reporting period.

Bibliography Type:	Description: (Last Updated: 06/13/2025)
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